

Lecture 9: Types of Biases in Data-driven Technology

HUM 4441: Engineering Ethics for Computer Science Engineers

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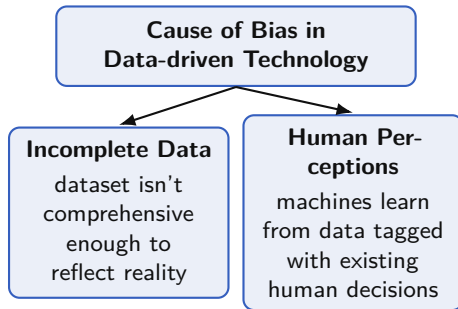
What Is Bias?

Bias

Prejudice in favor of or against one thing, person, or group compared with another. This preference influences understanding and outcomes in a way that **prevents neutrality or objectivity**.

Bias in Machine Learning

Occurs when the predictions or decisions produced by a model are **unfair** - a symptom of the data or the way the problem was framed, not a flaw in the math itself.

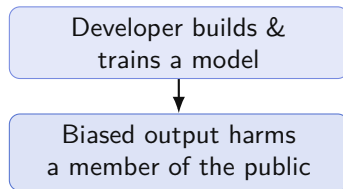


The Software Engineer's Duty

Recall Lecture 5: engineers hold a **professional duty to protect the public**. When the “product” is an algorithm, that duty extends to the people affected by its predictions.

ACM Code of Ethics

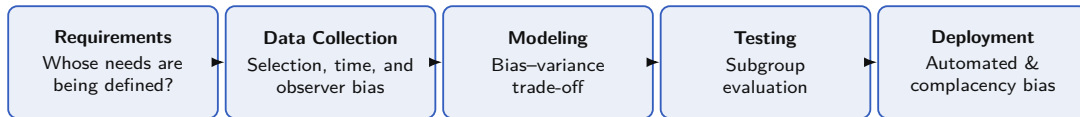
- **1.4** Be fair and take action not to discriminate
- **2.5** Give comprehensive evaluations of systems and their impacts, including possible risks
- **3.1** Articulate social responsibilities and encourage full participation



Recognizing bias is the first step toward preventing this outcome.

Bias Across the Software Development Lifecycle

Bias is not a single bug injected at one point - it can enter, and must be checked for, at **every stage** of building a data-driven system.



Sociological Bias: Society-Level Prejudice

1 Racism

Belief that one's own race or culture is superior (ethnocentrism).

"British drive on the wrong side of the road."

In practice: racial profiling in predictive-policing tools.

2 Sexism

Belief that one gender is superior.

"Men are stronger and do all the work."

In practice: a hiring model trained on male resumes penalizes women.

3 Ageism

Stereotyping based on age.

"Older people can't use new tech."

In practice: interfaces and health-tech that exclude older users.

4 Religious Prejudice

Irrational fear or hatred of a religion.

"Followers of religion X are dangerous."

In practice: discriminatory content moderation and surveillance systems.

Lecture 10, Types of Bias

Implicit Bias: The Unconscious Kind

Definition

Negative attitudes or stereotypes towards a person or group that **we aren't even consciously aware of**.

- **Not accessible through introspection** - revealed only through implicit association tests
- **May oppose our conscious beliefs** - a believer in equality can still hold implicit bias
- **Pervasive** - everyone possesses them to some degree
- **Caused by exposure (or lack of it)** - stereotypes absorbed from friends, family, or limited knowledge of a group



Cognitive Bias: Mental Shortcuts

1 Anchoring

Over-reliance on the **first** piece of information offered.

Doctors diagnosing based on first impressions.

2 False Consensus

The tendency to **over-estimate** how normal our own values and ideas are.

3 Optimism Bias

Believing our own chances of experiencing something negative are **lower** than those around us.

4 Attribution Error

Explaining others' behavior by **character**, not circumstance.

"I was late because of traffic."



"It's because you are lazy."

4 Problems That Biases Help Us Address

Cognitive biases are not just flaws - they're mental shortcuts our brains evolved to survive an overwhelming amount of information.

1 Too Much Information

There is too much information in the world - we filter almost all of it out.

2 Not Enough Meaning

We see only a tiny part of the world but must make sense of it to survive.

3 What to Remember?

We can only keep information that is likely to prove useful in the future.

4 Need to Act Fast

We're constrained by time and information, and cannot let that paralyze us.

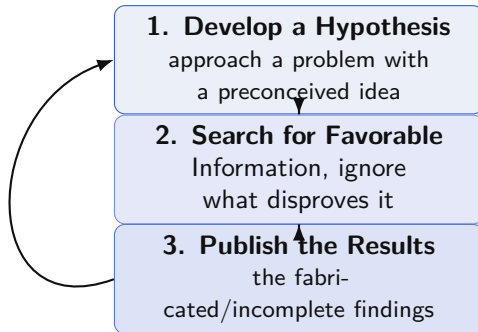
The same shortcuts that help a person survive daily life can quietly steer a **requirements meeting** or a **data-labeling decision** - we need to be aware of such.

Confirmation Bias

Definition

The tendency to search for, interpret, favor, and recall information that **confirms** one's pre-existing beliefs.

In data science, this can produce models that confirm our own assumptions rather than discover the true patterns - e.g., collecting data that validates a feature you already believe is important.



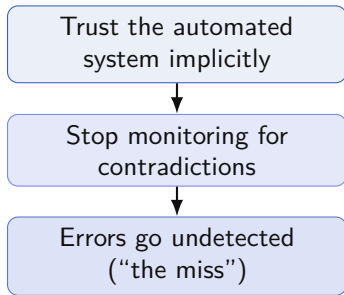
Complacency Bias & Automated Bias

Complacency Bias

People **trust automated systems over human decisions** because they are perceived as reliable and objective.

Automated Bias

People **disregard** the search for contradictory information, feeling the system is so **reliable** it does not require monitoring - leading to **omission errors**: the machine fails to detect a problem, and nobody notices.



Example: Spell Check, GPS - we often accept the correction or route even when it's wrong.

Data Collection Bias

1 Selection

Not randomized; data collection isn't representative.

Selecting one demographic to predict a vote.

2 Time Interval

A single time period is selected instead of a wide range.

Estimating profit from one unusually good month.

3 Observer

People judge with cognitive bias and are not objective.

A satisfaction survey sent right after a workshop.

4 Misclassification

Data is incorrectly classified or labeled.

Mislabeled data quietly rigs the final result.

Lecture 10, Types of Bias

Statistical Bias

1 Outliers

Data points far from the rest can be removed from a dataset.

25, 29, 32,
110, 33, 27, 28

2 Underfit

Removing variables might cause the model to **underfit** - important information is left out.

3 Avoid Overestimating

Include as many variables as possible **before** judging what to exclude.

4 Remove Uncorrelated

Remove data that is uncorrelated.

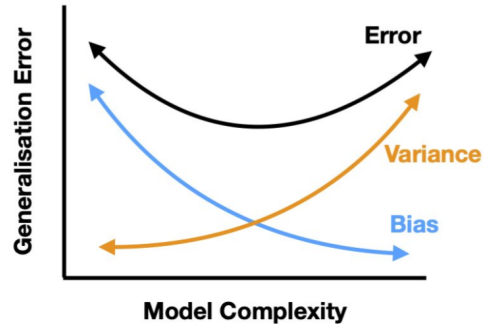
Drowning deaths are not caused by ice cream sales.

Modeling Bias: The Bias-Variance Trade-off

- **Bias:** how well a model matches a training set. A model with **low bias** aligns closely to the training set.
- **Variance:** how much a model changes when you use different portions of your dataset during training. A model with **high variance** is very flexible.
- **Underfit models:** high bias, low variance.
- **Overfit models:** low bias, high variance.

The Goal

We want **both low bias and low variance** for a well-trained model.



Fairness Metrics: Can We Even Define “Fair”?

Once bias is found, engineers still must choose **how to measure fairness** - and different definitions can conflict with each other mathematically.

Demographic Parity

Positive outcome rates are equal across groups, regardless of actual qualification.

Equalized Odds

Error rates (false positive / false negative) are equal across groups.

The Impossibility Result

Except in special cases, a model **cannot simultaneously satisfy** both demographic parity and equalized odds when base rates differ between groups.

Engineering implication: choosing a fairness metric is itself an **ethical decision**, not a purely technical one - and it should be documented and justified, not left as a silent default.

An Engineer's Responsibility: Mitigating Bias

1 Audit the Data

Check who is - and isn't - represented before training. Document dataset limits, as with any test report.

2 Test Across Subgroups

Evaluate accuracy separately for race, gender, and age - not just one overall average.

3 Diversify the Team

Diverse teams are less prone to affinity bias and more likely to catch blind spots early.

4 Document & Disclose

Publish known limitations and risks - this is **conscientious** engineering, not a marketing decision.

5 Don't Automate Blindly

Keep a human in the loop. Guard against complacency and automated bias in deployment.

6 Speak Up

If management ignores a known bias risk, recall Lecture 5: this may be grounds for **whistle-blowing**.

Legal & Professional Accountability

Regulation Is Catching Up

- **EU AI Act (2024):** risk-based obligations; high-risk systems (hiring, credit, policing) need bias testing and documentation
- **GDPR Art. 22:** right to an explanation for automated decisions with legal effect
- **IEEE 7000 series:** process standards for embedding ethics into system design

So What, as a CS Engineer?

- “The client didn’t ask for it” is not a defense once bias testing is a known best practice
- Professional codes (ACM/IEEE) can be cited in liability and disciplinary proceedings
- Compliance is a **floor**, not a ceiling - ethical practice usually exceeds the legal minimum

Case Study: Bias in Face Detection

The Problem

Many early facial recognition systems were trained predominantly on **white, male faces** - accurate for that group, but performing far worse on women and people of color.

Algorithmic Accountability

Joy Buolamwini's MIT Media Lab *Gender Shades* study exposed significant bias in commercial systems from IBM, Microsoft, and Amazon.

Real-World Impact

- Law enforcement misidentifying suspects
- Photo-tagging features failing certain users

Selection Bias + Modeling Bias + Sociological Bias
→ converge into real-world unfairness

Case Study: The COMPAS Recidivism Algorithm

The System

COMPAS scored defendants' risk of reoffending, informing U.S. judges' bail and sentencing decisions.

The Investigation

A 2016 ProPublica analysis found Black defendants were far more likely to be **falsely flagged as high-risk**, and white defendants more likely to be **falsely flagged as low-risk**.

The Hard Ethical Question

The vendor countered that the tool was **equally accurate overall** for both groups - illustrating the fairness-metric conflict from two slides ago: equal accuracy does not mean equal error types.

A trade secret algorithm shaping a person's liberty, with no public audit trail, remains one of the most cited cases in engineering-ethics discussions of automated bias.

Summary

Human Origins

- Sociological bias
- Implicit / affinity bias

Individual Thinking

- Anchoring, false consensus, optimism, attribution error
- Confirmation bias

The Data Pipeline

- Selection, time interval, observer, misclassification
- Outliers, uncorrelated data

The Model & The Engineer

- Complacency & automated bias
- Bias-variance trade-off
- Audit, test, disclose

Bias enters at every stage - and the engineer is professionally responsible for catching it.

Homework: Case Study on Face Detection

Task

- Research a real-world case of **bias in facial recognition or detection**
- Identify which types of bias (sociological, selection, modeling) were involved

Question

Which article(s) of the **ACM Code of Ethics** were most relevant to how this bias could have been prevented, and at which stage of the pipeline should the engineers have intervened?

Thank You